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Abstract

The recent outbreaks of COVID-19 have highlighted the demand for the rapid development and deployment of PCR based pathogen detection kits. This requires careful design of specific PCR primers that are optimized for melting temperature, GC content, and avoiding primer dimerization. For reactions with only a few primers, this challenge can be easily managed by a laboratory technician. However, with a large number of primer pairs (a multiplex PCR), the number of possible interactions becomes too difficult for a human to fully optimize. Machine learning is a better-suited process for dealing with these challenges.

Our research aims to use standard supervised machine learning approaches including L1 regularized regression (LASSO) to better understand the features that impact multiplex PCR performance. Our data set includes the results of 18 different primer sets (with up to one hundred primers each) on a total of 138 samples. We are training our model using individual primer features and primer interactions to learn a function to predict primer set performance. We will compare our final model to existing heuristic multiplex PCR design approaches. If our machine learning approach is successful, it will vastly improve the design and reduce deployment time for pathogen detection assays for future outbreaks.

Background

- A Polymerase chain reaction (PCR) is a technique used to synthesize many copies of a specific region of DNA. Primers are strands of nucleic acid used for DNA synthesis. These primers are designed to target and amplify a region of interest in the template DNA.

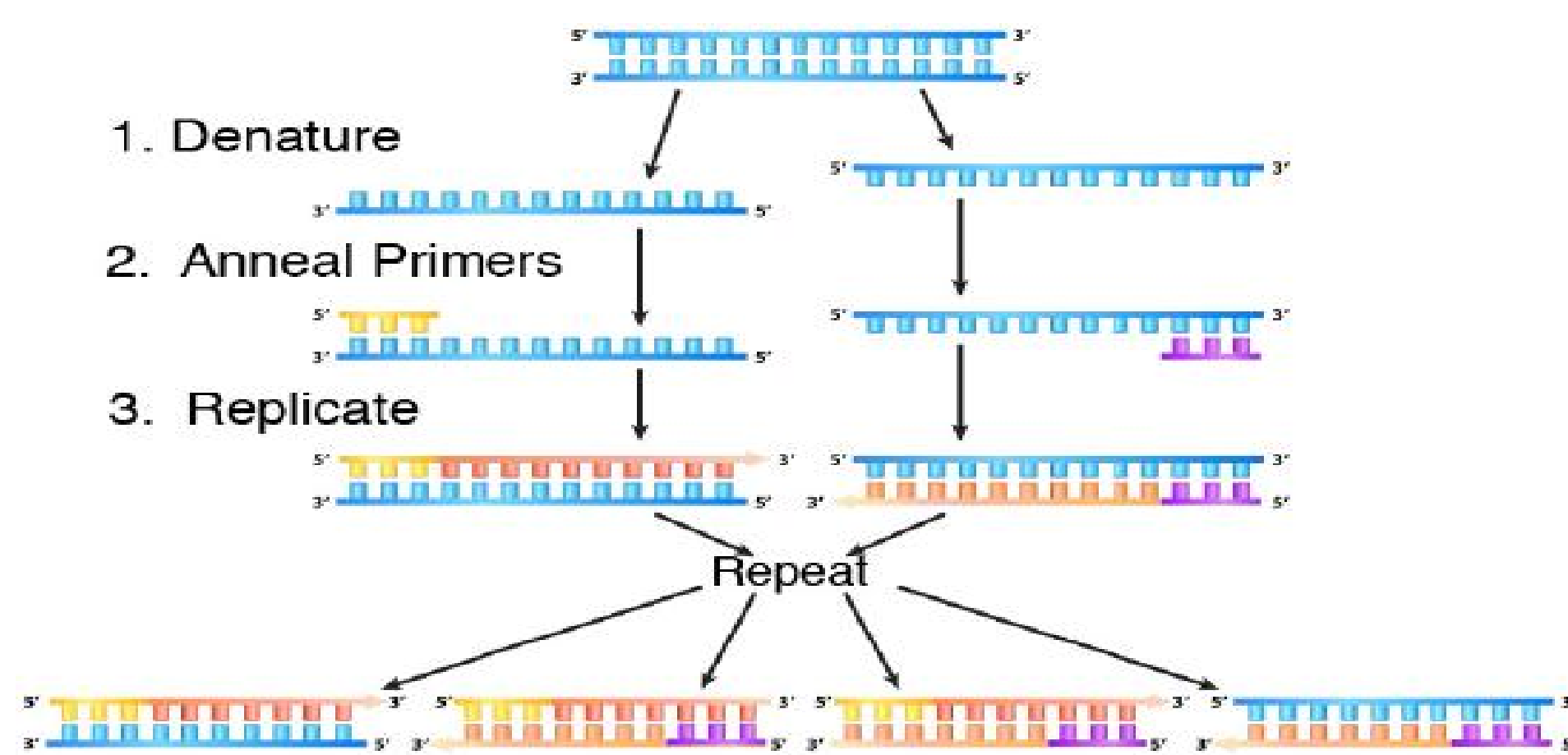


Fig. 1: A Standard PCR

- It is often valuable to amplify multiple target regions in one PCR. To do this, we use a multiple primer in one PCR (a multiplex PCR).
- Due the cross interaction between primers, optimizing primer design for a multiplex PCR can be difficult for humans to do themselves. With our data including information about primer features such as GC content, melting temperature, and primer length, we can use machine learning algorithms to analyze the success of these features (and more) in a primer set.

Support

This work was supported by ... NAU Interns to Scholars

Data Exploration

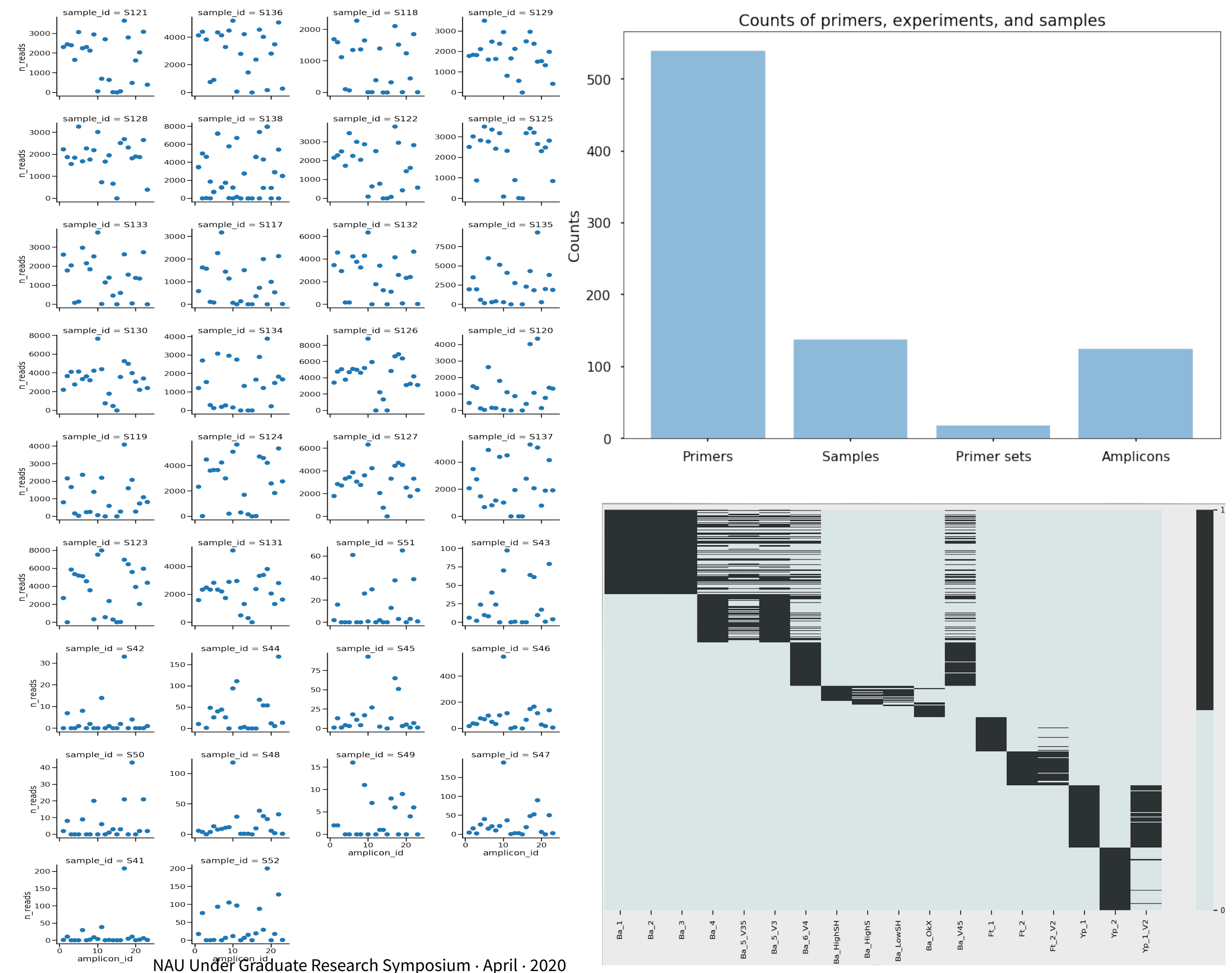


Fig. 2: Data Exploration

We started by creating a figure that gives us the count of each important aspect of our data. Our data set contained 512 different primers, 138 samples, 18 primer sets, and 132 amplicons. Some of these primers can be spread across multiple primer sets, so we created a binary heat map which would show this overlap.

Future Analysis

With our data fully explored, we now plan on implementing machine learning algorithms to optimize primer design. First, we plan on using k-fold cross validation on our data set. Once this is done, we plan on applying different types of machine learning algorithms to the data so that we can get some learned models.

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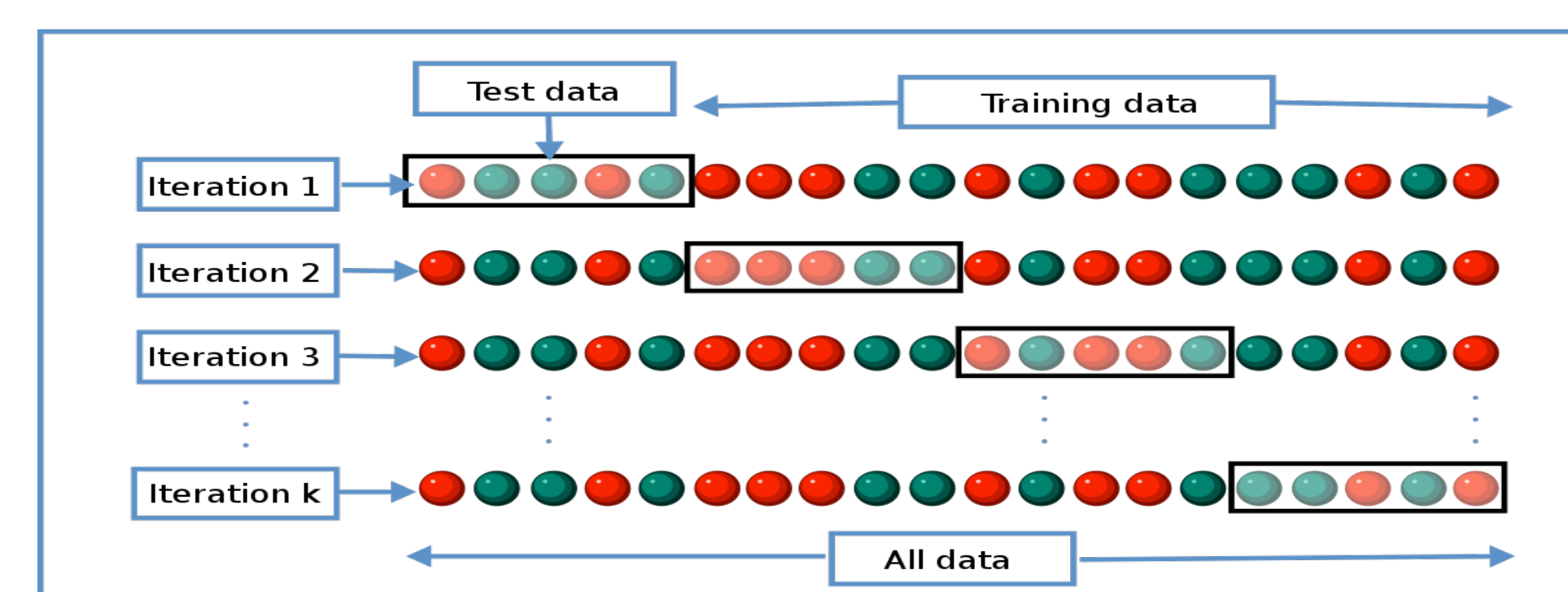


Fig. 3: K-Fold CV Figure

- Once the the K-Fold CV is complete, we plan on running our data through different machine learning models. Some of the models we plan on trying include L1-regularized poisson/binomial regression, decision tree, random forests, boosting, and neural networks.